Hydrogenic Bound States from Phase Holonomy Part II

A Derivation within Super Information Theory

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Abstract

In standard quantum mechanics, hydrogenic structure is obtained by solving the Schrödinger equation in a postulated Coulomb potential. In Super Information Theory (SIT), binding and state discreteness arise instead from the dynamics and topology of two primitive fields: a real time-density scalar $\rho_t(x)$ and a complex coherence field $\psi(x) = R_{\text{coh}}(x)e^{i\theta(x)}$. Modeling the proton as a localized coherence defect produces a static, spherically symmetric perturbation of the time-density field with vacuum Green-function profile $\rho_t(r) \simeq \rho_0(1 - \kappa/r)$ outside the core. When substituted into the SIT informational energy density, this profile yields an emergent inverse-square attraction, $F(r) \propto -1/r^2$ for $r \gg \kappa$, reproducing Coulomb scaling without taking electric charge as a primitive input. Stability and discrete bound configurations arise from the coherence-sector gradient energy, while allowed states are selected by a global holonomy constraint enforcing single-valued coherence phase. In this framework, hydrogenic shells correspond to discrete geometric sectors determined by informational dynamics and topological consistency.

1 Introduction: From Probability to Informational Topology

The stability of matter remains one of the most fundamental questions in physics. In the standard quantum-mechanical framework, the hydrogen atom is described as a probabilistic system in which the electron occupies stationary eigenstates of the Schrödinger equation within a postulated electrostatic potential $V(r) \propto -1/r$. This framework is extraordinarily predictive and experimentally successful. At the same time, it treats both the Coulomb potential and the discreteness of allowed states as axiomatic inputs. As a result, it does not, by itself, address why the binding interaction exists in the first place, nor why only discrete, stable configurations persist.

Super Information Theory (SIT) approaches this question from the principle of *Informational Realism*, which proposes that mass, force, and stability are not taken as primitive ingredients, but arise as emergent features of underlying informational fields. Within this framework, the fundamental substrate of physics consists of two coupled fields: a real *time-density* scalar $\rho_t(x)$, representing the local rate of causal updates, and a complex *coherence field* $\psi(x) = R_{\text{coh}}(x)e^{i\theta(x)}$, representing the magnitude and phase of informational order. Localized, long-lived particles are described as stable excitations of these fields rather than as irreducible point objects.

Within SIT, the electron is therefore not assumed as a fundamental input but appears as an emergent, coherence-supported excitation. Its persistence as a massive entity is governed by

energetic and topological constraints on the underlying fields rather than by externally imposed quantum numbers. The primary aim of this paper is to show that, once the SIT field content and energy functional are specified, the qualitative structure associated with the hydrogen atom follows without being separately postulated.

In this limited but precise sense, the present work establishes a *necessity result* in theoretical physics: given the SIT action and symmetry principles, inverse-square attraction, repulsive stability, and discrete bound configurations are not optional assumptions but unavoidable consequences. No continuous family of bound solutions is consistent with the coherence dynamics assumed by SIT.

The argument is not intended to replace quantum electrodynamics within its established domain of validity. Rather, it shows that hydrogenic structure emerges as the only self-consistent and dynamically stable outcome of the informational dynamics posited by SIT.

We demonstrate that, within the SIT framework, discrete hydrogenic shells are enforced by topological and coherence constraints, and that no continuous bound configurations are dynamically admissible.

Before turning to the detailed derivation, it is important to emphasize that inverse-square attraction alone is not sufficient to account for atomic stability within SIT. The $1/r$ behavior arises generically as the vacuum Green-function solution of the static time-density field sourced by a localized coherence defect, while stability against collapse is supplied by the coherence sector. In particular, the phase-gradient contribution $R_{\mathrm{coh}}^2(\nabla\theta - qA)^2$ provides a repulsive energetic barrier that penalizes phase mismatch and establishes finite equilibrium configurations. Discrete bound states are further selected by a global single-valuedness (holonomy) constraint on the coherence phase, which restricts admissible configurations to topologically consistent phase histories. Although these structures admit an interpretation in which electromagnetic interactions emerge as effective descriptions of underlying informational and topological constraints, this interpretive layer is not required for the internal consistency of the derivation. Continuity with standard atomic physics is addressed separately, where the decohered QFT limit of SIT is shown to recover a locally $U(1)$ gauge-invariant electromagnetic sector and the conventional hydrogenic spectrum. A complementary manuscript, *Hydrogenic Bound States from Phase Holonomy Part I*, addresses the decohered QFT limit of the framework and shows how a locally $U(1)$ gauge-invariant electromagnetic sector and the standard hydrogenic spectrum are recovered.

1.1 The Emergence of Mass and Attraction

In conventional physics, mass m is treated as an intrinsic property of particles. SIT instead relates mass to localized informational energy. The informational energy density of a spacetime region is given by the SIT Master Energy Equation,

$$E(x) = \zeta\, R_{\mathrm{coh}}(x)\, [\rho_t(x)]^2, \tag{1}$$

where ζ is the Informational Inertia Constant. The rest mass associated with a stable excitation is correspondingly defined as

$$m(x) = \frac{\zeta}{c^2} R_{\mathrm{coh}}(x)\, [\rho_t(x)]^2. \tag{2}$$

To model hydrogen, we treat the proton as a localized, high-coherence defect that perturbs the surrounding vacuum configuration of the time-density field. The static, spherically symmetric solution of the associated scalar field equation yields the asymptotic profile

$$\rho_t(r) \simeq \rho_0 \left(1 - \frac{\kappa}{r}\right), \tag{3}$$

outside the nuclear core. Substituting this profile into the informational energy density and evaluating the resulting spatial gradient produces an effective inverse-square attractive force,

$$F(r) \propto -\frac{1}{r^2}, \tag{4}$$

recovering the characteristic scaling of the Coulomb interaction. In this derivational sense, attraction arises from the geometry of the time-density field.

1.2 Stability, Repulsion, and Holonomy Constraints

The kinetic term of the coherence field contains a positive-definite phase-gradient contribution,

$$E_{\text{phase}} \propto R_{\text{coh}}^2 (\nabla\theta - qA)^2, \tag{5}$$

which penalizes phase mismatch. Configurations that force incompatible phase alignment generate large gradients in θ, producing a rapidly increasing energetic cost. This mechanism supplies a natural repulsive contribution that prevents collapse and suppresses incoherent configurations.

For a closed trajectory γ, the coherence phase must be single-valued, enforcing the holonomy constraint

$$\oint_\gamma (\nabla\theta - qA) \cdot d\ell = 2\pi n, \qquad n \in \mathbb{Z}. \tag{6}$$

If this condition is violated, the phase-gradient energy diverges and the coherence amplitude R_{coh} is suppressed, destabilizing the excitation. The discrete shells associated with the hydrogen atom thus correspond to the only geometrically and energetically admissible configurations in which time-density attraction and coherence constraints are simultaneously satisfied.

Electromagnetic structure may be viewed as a secondary, effective description of these underlying informational and topological constraints. However, this interpretive layer is not required for the consistency of the derivation presented here and remains subject to further theoretical and empirical refinement.

2 Topological Necessity of Hydrogenic Shells: A Geometric Derivation

Standard quantum mechanics provides an algebraic derivation of hydrogenic structure: assuming a Hamiltonian with a Coulomb potential and a wavefunction defined on Hilbert space, the discrete spectrum follows as the set of normalizable eigenvalues of the Schrödinger or Dirac equation. In that formulation, discreteness is encoded spectrally.

The derivation presented here is geometric and topological in character. The axioms differ: reality is modeled in terms of a real time-density field $\rho_t(x)$ and a complex coherence field $\psi = R_{\text{coh}} e^{i\theta}$. Within this framework, continuous families of bound configurations are dynamically unstable. Quantization instead emerges as a condition for the long-term persistence of coherent informational structure, rather than as an imposed eigenvalue rule.[1]

Under the axioms of Super Information Theory (SIT), hydrogenic bound states arise as unavoidable consequences of coherence dynamics and gauge structure. The argument is constructive: once the coherence field and its coupling to the $U(1)$ connection are specified, only discrete topological sectors support stable bound configurations.

[1] This statement concerns stability within the SIT dynamical framework and does not assert inconsistency of alternative formulations.

1. Recovery of Attractive Binding via Gauge Holonomy

The binding interaction is not postulated as an independent force. As shown in Super Information Theory (v78, Sec. 6.5), the $U(1)$ gauge sector emerges from the holonomy structure of the coherence phase. Varying the SIT master action with respect to the connection A_μ recovers the Maxwell equations.

For a static localized source (the proton), the resulting field satisfies Gauss' law, yielding the Coulomb potential $\phi(r) \propto 1/r$. This reproduces the familiar inverse-square attractive interaction through the vector sector.[2]

2. Repulsion via Decoherence: The Phase-Gradient Energy Barrier

Attraction alone does not guarantee stability. In SIT, the informational energy density contains a positive-definite phase-gradient contribution,

$$E_{\text{grad}} \propto R_{\text{coh}}^2 (\nabla\theta - qA)^2.$$

This term should be interpreted as an energetic cost associated with phase mismatch, not as a classical repulsive force.[3]

When a coherence-supported excitation attempts to occupy a configuration incompatible with the ambient phase geometry—such as a non-resonant inward spiral—the phase gradient $\nabla\theta$ increases sharply. The associated energy cost grows without bound, rendering such configurations dynamically unstable.

Decoherence is therefore not treated as purely stochastic noise, but as a dynamical suppression mechanism: configurations that violate global phase consistency become energetically unfavorable, leading to the attenuation of R_{coh}.

3. Quantization as a Topological Stability Condition

A long-lived excitation must minimize phase-gradient energy while remaining bound by the vector interaction. This balance occurs only when the coherence phase is globally single-valued. The admissibility condition is the holonomy constraint

$$\oint_\gamma (\nabla\theta - qA) \cdot d\ell = 2\pi n, \qquad n \in \mathbb{Z}.$$

For integer n, phase transport is globally consistent, coherence is maintained, and the excitation remains dynamically stable ($m \propto R_{\text{coh}}$). For non-integer values, phase mismatch generates large gradients, the gradient energy diverges, and coherence rapidly decays.

In this sense, discrete hydrogenic shells correspond to the only topological sectors that support persistent coherence-supported bound states within the SIT framework.[4]

[2] This recovery does not imply that electromagnetism is reduced to topology alone; it arises from the combined gauge and coherence structure assumed by the theory.

[3] No additional fundamental force is introduced here; the effect follows from the energy functional governing coherence.

[4] This does not claim that bound states are impossible in other theoretical frameworks, only that they are restricted in this one.

4. Scope and Interpretation of the Result

Within the axioms of Super Information Theory, the derivation establishes the following:

- **Structural necessity:** Continuous bound configurations are dynamically unstable; only discrete topological sectors persist.
- **Enforcement mechanism:** Discreteness arises from the divergence of phase-gradient energy under decoherence, not from imposed quantization rules.
- **Compatibility:** The resulting structure reproduces standard hydrogenic phenomenology while arising from a distinct informational geometry.

Falsifiability and limits. These conclusions are conditional on the SIT axioms, particularly the form of the coherence energy functional and its coupling to gauge structure. Any empirical or theoretical demonstration that stable, coherence-preserving bound states exist in violation of the holonomy constraint would falsify this derivation. The argument is therefore interpretive and structural rather than predictive, aimed at explaining why quantization is required for stability once the SIT framework is assumed.

3 Scope and Limitations

The results presented in this work are intended to establish a specific and limited claim: that, within the internal assumptions of Super Information Theory (SIT), the qualitative structure associated with the hydrogen atom can be derived rather than postulated. In particular, the analysis demonstrates how inverse-square attraction, dynamical stability, and discrete bound configurations emerge from the coupled dynamics of the time-density and coherence fields.

Several important limitations should be emphasized. First, the present treatment is restricted to static, spherically symmetric configurations and does not address dynamical processes such as radiation, scattering, or time-dependent transitions between bound states. Second, the analysis focuses on scaling behavior and topological constraints rather than precise numerical agreement with spectroscopic data. Recovering exact energy levels, fine structure, and relativistic corrections would require additional structure beyond the minimal framework employed here.

Third, while the formalism recovers features commonly associated with electromagnetism, the present work does not claim to derive the full content of quantum electrodynamics. The appearance of Coulomb-like behavior should be understood as an effective result within the specific regime considered, rather than as a replacement for established gauge theories in their full domain of applicability.

Finally, interpretive statements regarding the foundational origin of mass, charge, or electromagnetic structure are explicitly secondary to the derivational results. Such interpretations are not required for the internal consistency of the analysis and remain subject to further theoretical development and empirical constraint.

Within these bounds, the aim of this paper is to demonstrate logical sufficiency rather than empirical completeness: given the SIT field content and energy functional, hydrogenic structure is not optional but follows as a constrained outcome.

Relation to Effective Field Theories

The perspective adopted here is compatible with the general philosophy of effective field theories. In this view, standard quantum mechanics and quantum electrodynamics may be understood as

effective descriptions that accurately capture observable behavior within a restricted regime, without specifying the deeper origin of their parameters or interaction terms. The SIT framework does not seek to invalidate such descriptions, but instead explores whether features typically encoded as effective inputs—such as inverse-square attraction and discrete bound states—can arise from a more primitive informational substrate. Where standard theories succeed empirically, they are recovered in SIT as limiting or emergent descriptions appropriate to the scales considered.

4 Theoretical Framework: The SIT Master Equations

Super Information Theory (SIT) is founded on the principle of *Informational Realism*: the hypothesis that the fundamental substrate of physical reality is not matter or spacetime itself, but information encoded in dynamical fields. In this view, observable quantities such as mass, force, and stability arise as emergent manifestations of informational structure and its evolution, rather than as primitive inputs.

To formalize this framework, SIT introduces a minimal set of fields that quantify both the degree of informational order and the rate at which information is processed locally. These fields are defined at every point in spacetime and constitute the primitive variables from which all subsequent derivations follow.

4.1 The Two-Field Basis of Coherence

The informational state of spacetime in SIT is specified by two coupled, gauge-invariant scalar fields:

- **Coherence Ratio** $R_{\text{coh}}(x)$: a dimensionless scalar field, $0 \leq R_{\text{coh}} \leq 1$, representing the normalized density of mutual information or phase alignment within a spacetime region. Physically, R_{coh} quantifies the degree of structured correlation and serves as a measure of local informational order.
- **Time-Density** $\rho_t(x)$: a real scalar field with units of inverse time, $[\rho_t] = \mathrm{T}^{-1}$, representing the local rate of informational state updates. The time-density governs the effective dynamical scale of processes and plays the role of a locally variable causal update rate.

Together, R_{coh} and ρ_t form the minimal informational basis required to describe stable excitations, dynamical interactions, and energetic gradients within SIT.

4.2 The Principle of Informational Energy Equivalence

In conventional physics, energy is commonly treated as a property carried by mass, as expressed by $E = mc^2$. SIT inverts this perspective through the *Principle of Informational Energy Equivalence*, which posits that energy is a derived quantity arising from informational order and update rate.

This principle is encoded in the SIT Master Energy-Density Equation,

$$E(x) = \zeta\, R_{\text{coh}}(x)\, [\rho_t(x)]^2, \tag{7}$$

where:

- $E(x)$ is the local informational energy density,

- ζ is the *Informational Inertia Constant*, a fundamental proportionality factor with dimensions $[\zeta] = \mathrm{ML}^2$, analogous to a moment of inertia, which relates informational dynamics to physical resistance.

This relation implies that energy scales linearly with informational order and quadratically with the rate of causal updates. Conceptually, the fixed conversion factor c^2 in relativistic physics is replaced by the dynamically varying field ρ_t^2, allowing energy to respond to local informational geometry.

4.3 Emergent Mass as Localized Informational Energy

A direct consequence of the Master Energy-Density Equation is that mass is not an intrinsic attribute of matter, but rather the manifestation of stable, localized informational energy. By identifying rest energy density with the SIT energy density, we define the emergent rest mass associated with a localized excitation as

$$m(x) = \frac{\zeta}{c^2} R_{\mathrm{coh}}(x) \, [\rho_t(x)]^2. \tag{8}$$

In this framework, a "massive" particle such as an electron corresponds to a long-lived, coherence-supported excitation characterized by nonzero intrinsic R_{coh} and a stable ρ_t profile. Conversely, massless excitations correspond to propagating modes for which the effective rest-frame coherence approaches zero.

This definition provides the foundation for the subsequent analysis: once mass is understood as localized informational energy, the response of such an excitation to gradients in the time-density field can be analyzed directly from the SIT energy functional, without introducing external force postulates.

5 The Emergence of Inverse-Square Attraction

To derive the structure of the hydrogen atom within Super Information Theory, we first establish the origin of the binding interaction between an electron-like excitation and the nucleus. In the Standard Model, this interaction is introduced axiomatically as the electrostatic Coulomb force,

$$F(r) = -k_e \, \frac{q_1 q_2}{r^2}.$$

In contrast, SIT does not take electric charge as a fundamental primitive at the level of informational geometry. Instead, long-range attraction arises as a consequence of gradients in the time-density field that determine the local energetic cost of sustaining a coherent excitation.

5.1 The Nucleus as a Time-Density Well

We model the atomic nucleus (proton) as a localized, high-coherence defect embedded in the informational vacuum. Such a defect perturbs the surrounding equilibrium configuration of the time-density field $\rho_t(x)$, producing a static, spherically symmetric distortion. Physically, this distortion corresponds to a local modification of the rate of causal updates, analogous in form to gravitational time dilation near a compact source.

Outside the nuclear core, where the defect can be treated as pointlike, the static field equation for ρ_t admits the vacuum Green-function solution

$$\rho_t(r) \simeq \rho_0 \Big(1 - \frac{\kappa}{r} \Big) , \tag{9}$$

where ρ_0 is the asymptotic vacuum time-density and κ characterizes the effective coherence scale of the nucleus. This profile is the unique spherically symmetric harmonic solution consistent with a localized source.

5.2 Derivation of the Inverse-Square Attractive Force

The interaction experienced by a coherence-supported excitation is determined by the local informational energy density. Substituting the time-density profile Eq. (9) into the SIT Master Energy-Density Equation yields

$$E(r) = \zeta\, R_{\text{coh}}\, [\rho_t(r)]^2 = \zeta\, R_{\text{coh}}\, \rho_0^2 \left(1 - \frac{\kappa}{r}\right)^2 . \tag{10}$$

Expanding the square gives

$$[\rho_t(r)]^2 = \rho_0^2 \left(1 - \frac{2\kappa}{r} + \frac{\kappa^2}{r^2}\right) . \tag{11}$$

At atomic length scales, the orbital radius satisfies $r \gg \kappa$, allowing the κ^2/r^2 term to be neglected. The leading-order energy density then simplifies to

$$E(r) \simeq \zeta\, R_{\text{coh}}\, \rho_0^2 \left(1 - \frac{2\kappa}{r}\right) . \tag{12}$$

The effective force acting on a localized excitation is obtained from the spatial gradient of the total informational energy. For radial symmetry, this reduces to

$$F_{\text{SIT}}(r) = -\frac{dE(r)}{dr} = -\,2\zeta\, R_{\text{coh}}\, \rho_0^2\, \frac{\kappa}{r^2}. \tag{13}$$

Thus,

$$F_{\text{SIT}}(r) \propto -\frac{1}{r^2}. \tag{14}$$

The negative sign indicates attraction toward the nucleus. The characteristic inverse-square scaling arises directly from the geometry of the time-density field and does not rely on the postulation of electric charge or photon exchange at the fundamental level.

5.3 Interpretation

Within this framework, long-range attraction is a geometric consequence of the informational energy landscape set by the time-density field. A coherent excitation minimizes its informational energy by moving toward regions of lower ρ_t, producing effective radial acceleration with inverse-square scaling.

This result identifies an attractive contribution consistent with hydrogenic binding within the SIT framework. However, attraction alone does not guarantee stability. As shown in the following section, collapse is prevented and discrete bound configurations are enforced by the positive-definite phase-gradient energy of the coherence field, which supplies a repulsive barrier and imposes topological constraints on admissible trajectories.

6 Topological Stability and Orbital Quantization

The existence of an attractive inverse-square interaction alone does not guarantee stable atomic structure. In classical electrodynamics, a charged particle subject to such a force would radiate energy and spiral into the nucleus. In Super Information Theory, stability is not imposed as an additional postulate; instead, it follows from energetic and topological constraints on the coherence field that forbid incoherent trajectories.

6.1 Coherence Conservation and Phase-Gradient Energy

In SIT, a localized excitation possesses rest mass only insofar as it maintains a nonzero coherence ratio. From the definition of emergent mass,

$$m(x) \propto R_{\mathrm{coh}}(x)\,[\rho_t(x)]^2,$$

it follows that configurations in which coherence is strongly degraded cannot sustain a long-lived massive excitation.

The coherence field $\psi = R_{\mathrm{coh}} e^{i\theta}$ contributes a positive-definite kinetic term to the informational energy functional,

$$E_{\mathrm{phase}} \propto \int R_{\mathrm{coh}}^2\,(\nabla\theta - qA)^2\,d^3x. \tag{15}$$

This term penalizes spatial phase mismatch. Any trajectory that forces incompatible phase alignment generates large phase gradients and incurs a rapidly increasing energetic cost. This mechanism, familiar from superfluids and Ginzburg–Landau theory, supplies a natural repulsive barrier that prevents collapse into the nucleus and excludes incoherent motion.

6.2 Holonomy of the Coherence Field

The phase stability of a coherence-supported excitation is governed by the geometry of the $U(1)$ coherence-phase bundle. For a closed trajectory γ, parallel transport of the phase is characterized by the holonomy (Wilson loop)

$$U[\gamma] = \exp\left(\frac{iq}{\hbar}\oint_\gamma A_\mu\,dx^\mu\right), \tag{16}$$

where A_μ is the connection associated with the coherence phase.

Equivalently, the accumulated phase around γ may be written as

$$\Phi_\gamma = \oint_\gamma (\nabla\theta - qA)\cdot d\ell. \tag{17}$$

This quantity is gauge invariant and measures the total phase winding of the coherence field along the closed path.

6.3 Topological Constraint and Quantization

For a physical excitation to be well-defined, the coherence field must be single-valued. This imposes the global topological constraint

$$\oint_\gamma (\nabla\theta - qA)\cdot d\ell = 2\pi n, \qquad n \in \mathbb{Z}. \tag{18}$$

If this condition is violated, the coherence phase fails to close consistently, forcing large phase gradients and driving the phase-gradient energy upward. Such configurations are energetically unstable and suppress the coherence ratio, rendering the excitation short-lived.

The discrete allowed values of n therefore label the only geometrically admissible long-lived bound configurations. Orbital quantization in SIT is thus a consequence of global phase consistency rather than an imposed quantum rule.

6.4 Semiclassical Radius Spectrum

To connect this topological condition with familiar hydrogenic structure, we consider the semiclassical limit in which the phase gradient is related to the canonical momentum via the de Broglie relation $p = \hbar\nabla\theta$. The holonomy constraint then becomes

$$\oint_\gamma p \cdot d\ell = nh. \tag{19}$$

For a circular orbit of radius r, this yields the quantization of angular momentum,

$$L = mvr = n\hbar. \tag{20}$$

Balancing the centripetal acceleration with the inverse-square attraction derived in Section 3,

$$\frac{mv^2}{r} = \frac{k_{\text{eff}}}{r^2},$$

and eliminating v using the angular-momentum condition gives the discrete radii

$$r_n = \frac{n^2\hbar^2}{mk_{\text{eff}}}. \tag{21}$$

This spectrum mirrors the Bohr radius formula, but its interpretation is fundamentally different. In SIT, the discrete shells $n = 1, 2, 3, \ldots$ are not arbitrary eigenvalues but the unique radii at which inverse-square attraction, phase-gradient stability, and global coherence constraints are simultaneously satisfied. Hydrogenic structure thus emerges as a topological necessity enforced by the informational geometry of the coherence field.

7 Discussion

The derivation presented above reinterprets the structure of the hydrogen atom at a foundational level. While the resulting radius spectrum agrees with the Bohr model and, in the appropriate limit, with the principal quantum numbers of the Schrödinger equation, the underlying explanatory mechanism is fundamentally different. In this section we clarify the conceptual implications of this shift and outline the broader consequences of the SIT framework.

7.1 From Eigenvalues to Topological Necessity

In standard quantum mechanics, the discrete energy levels of the hydrogen atom arise as eigenvalues of the Hamiltonian operator acting on a Hilbert space. Quantization follows from analytic boundary conditions imposed on the wavefunction at the origin and at spatial infinity. Within this formalism, the integer quantum numbers label allowed solutions but do not themselves possess direct physical meaning beyond classification.

In Super Information Theory, discreteness arises from global consistency conditions on the coherence field rather than from the spectral properties of a differential operator. The allowed bound configurations correspond to the only geometries in which the coherence phase can remain globally well-defined and energetically stable within a time-density gradient. The integer n appearing in the holonomy condition is therefore interpreted as a topological winding number of the coherence phase rather than as an abstract quantum label.

From this perspective, hydrogenic shells are not arbitrary eigenstates selected by measurement, but the unique topological sectors in which attraction, phase-gradient stability, and coherence conservation are simultaneously satisfied. The familiar quantum numbers emerge as bookkeeping devices for underlying topological structure in the informational substrate.

7.2 Energetic Exclusion of Non-Admissible Regions

Standard quantum mechanics characterizes regions between shells as having low or vanishing probability density for stationary states. In SIT, the exclusion of such regions admits a dynamical interpretation rooted in the energy functional of the coherence field.

Configurations that violate the holonomy condition

$$\oint_\gamma (\nabla\theta - qA) \cdot d\ell = 2\pi n$$

necessarily generate large phase gradients. Because the phase-gradient contribution to the informational energy is positive definite, such configurations incur a rapidly increasing energetic penalty. As a result, coherence-supported excitations cannot persist in these regions for extended durations.

Rather than interpreting this behavior as a literal disappearance of the particle, SIT identifies forbidden regions as dynamically unstable configurations in which coherence is suppressed and the excitation rapidly transitions toward a nearby admissible topological sector. In this view, quantum transitions correspond to relaxation between the only energetically and topologically allowed configurations, rather than to instantaneous jumps between abstract eigenstates.

7.3 Macroscopic Implications and Experimental Outlook

The SIT Master Energy-Density Equation,

$$E = \zeta\, R_{\text{coh}}\, \rho_t^2,$$

is not restricted to atomic systems. Because gravity couples universally to energy, this relation suggests that coherence may contribute to gravitational sourcing whenever R_{coh} departs significantly from incoherent background values.

This observation motivates a potential macroscopic signature of SIT: systems exhibiting large-scale coherence may produce small but systematic deviations in their effective gravitational influence compared to incoherent systems with identical constituent mass and energy. Platforms such as Bose–Einstein condensates, superconducting circuits, or phase-locked quantum fluids provide natural environments in which such effects could, in principle, be probed.

At present, these considerations should be regarded as phenomenological predictions rather than established consequences. Any such coherence-dependent gravitational effect must be consistent with existing precision tests of gravity and equivalence principles. Nonetheless, the SIT framework provides a concrete mechanism by which coherence could enter gravitational physics, offering a falsifiable avenue for experimental investigation.

7.4 Summary

Taken together, the results of this work show that hydrogenic structure need not be assumed as a primitive feature of quantum theory. Within Super Information Theory, inverse-square attraction, orbital stability, and quantization are recovered as outcomes of informational geometry and coherence topology. The standard quantum-mechanical description is recovered as an effective representation of these deeper constraints, while SIT supplies a physical interpretation for why only certain configurations of matter can exist.

8 Conclusion

In this work we have shown that, given the foundational postulates of Super Information Theory, the qualitative structure associated with the hydrogen atom follows as a constrained outcome, without requiring the Coulomb potential or orbital quantization to be introduced as independent axioms. By reformulating the problem in terms of informational geometry, the familiar features of atomic physics can be understood as arising from coherence dynamics and topological consistency, rather than being introduced as independent assumptions.

8.1 Summary of Results

First, we demonstrated that a localized, high-coherence defect representing the nucleus induces a static, spherically symmetric distortion of the time-density field. When this vacuum solution is substituted into the SIT Master Energy-Density Equation,

$$E(x) = \zeta\, R_{\mathrm{coh}}(x)\, [\rho_t(x)]^2,$$

the resulting informational energy landscape generates an attractive force with inverse- square scaling. This establishes that the long-range binding interaction required for hydrogenic structure follows directly from gradients in the time-density field and does not need to be postulated independently.

Second, we showed that attraction alone is insufficient to guarantee stability. The positive-definite phase-gradient energy of the coherence field supplies a natural repulsive barrier that suppresses incoherent configurations and prevents collapse. Long-lived bound states are selected by a global topological constraint: the coherence phase must be single-valued along closed trajectories. This requirement enforces an integer winding condition,

$$\oint_\gamma (\nabla\theta - qA) \cdot d\ell = 2\pi n, \qquad n \in \mathbb{Z},$$

which restricts admissible configurations to a discrete set. In the semiclassical limit, these constraints reproduce the familiar hydrogenic radius spectrum, while providing a physical interpretation for the origin of orbital quantization.

8.2 Final Remarks and Outlook

The results presented here suggest that key structural features of atomic physics need not be taken as primitive postulates, but can instead be understood as emergent consequences of a deeper informational substrate. Within Super Information Theory, stable matter exists only where coherence, energetic gradients, and global phase consistency can be maintained simultaneously. Quantum

numbers acquire a natural interpretation as labels of admissible topological sectors rather than as abstract eigenvalue indices.

At the same time, this work does not seek to replace the successful calculational framework of quantum electrodynamics. In regimes where coherence gradients are negligible, SIT reduces to the standard Dirac–Maxwell description, and all conventional predictions are recovered. The value of the present approach lies in providing an explanatory foundation for why such structures arise and in identifying controlled avenues for potential deviations in highly coherent or strongly inhomogeneous environments.

More broadly, these results support the central claim of Informational Realism: that mass, force, and stability are manifestations of structured informational dynamics. The hydrogen atom serves as a concrete demonstration that familiar physical laws can be recovered from coherence geometry, offering a unified perspective on the origin of matter and its organization.

A Notation-Level Comparison with Standard Quantum Mechanics

This appendix provides a limited, notation-level comparison between elements of Super Information Theory (SIT) and standard nonrelativistic quantum mechanics (QM). Its purpose is strictly to aid orientation for readers familiar with conventional formulations. No claim of conceptual or foundational equivalence is implied.

In standard QM, the state of a particle is described by a complex wavefunction $\Psi(\mathbf{x}, t)$, whose modulus squared $|\Psi|^2$ is interpreted probabilistically. Dynamics are governed by the Schrödinger equation with a prescribed Hamiltonian operator. In SIT, by contrast, the fundamental objects are physical informational fields: a real time-density scalar $\rho_t(x)$ and a complex coherence field $\psi(x) = R_{\text{coh}}(x)e^{i\theta(x)}$. The coherence amplitude R_{coh} plays a role analogous to wavefunction magnitude, while the phase θ governs dynamical and topological constraints.

State description. In QM, the wavefunction Ψ encodes the complete state description. In SIT, no single object plays this role; instead, physical configurations are determined by coupled field profiles $\{\rho_t, R_{\text{coh}}, \theta\}$ that minimize the informational energy functional subject to topological constraints.

Potential and interaction. The Coulomb potential $V(r) \propto -1/r$ is an external input in QM. In SIT, inverse-square attraction arises effectively from spatial variation of the time-density field sourced by a localized coherence defect. The resulting force law matches the Coulomb scaling in the appropriate regime without assuming electric charge as a primitive quantity.

Quantization. Discrete energy levels in QM arise from boundary conditions imposed on solutions of the Schrödinger equation. In SIT, discrete bound configurations arise from global holonomy constraints on the coherence phase,

$$\oint_\gamma (\nabla\theta - qA) \cdot d\ell = 2\pi n,$$

which restrict admissible configurations to topologically consistent sectors.

Interpretation. Probabilistic interpretation is central to standard QM. SIT does not assume a probabilistic ontology at the fundamental level; instead, probabilities may emerge as effective descriptions of underlying informational dynamics in appropriate limits.

This comparison is intended solely as a translation aid. The derivations in the main text do not rely on quantum-mechanical postulates, nor do they assume the Schrödinger equation as a starting point.

References

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